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LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Student Club Event Ticketing & Budget Portal

Architecture Selection

We chose Layered Architecture for the Student Club Event Ticketing & Budget Portal.

Architectural Choice:

The system is organized into a presentation layer, business/application layer, and data layer. The presentation layer handles interactions with Club Leads and Campus Admins, while the application components manage event proposals, approvals, ticket orders, payments and QR tickets.

Reason 1:

Layered Architecture provides clear separation of responsibilities between the user interface, event and approval processing, ticket/payment services, and stored data. This makes the portal easier to understand, maintain and modify.

Reason 2:

The architecture suits the centralized nature of the university portal, where event proposals and budget approvals follow a defined workflow from Faculty Coordinator to Finance Office and Dean, while ticket and payment operations are handled by dedicated components.

Security Advantage:

Separating payment and approval functionality into dedicated application components limits direct access to sensitive operations from the user interface and allows authorization checks to be applied at the appropriate service boundaries.

Performance Benefit:

Separating frequently used operations such as ticket ordering and QR-ticket validation into dedicated components allows these operations to be processed without unnecessarily involving unrelated proposal and approval functionality.

Diagram:

